

Problem Set 1 with Suggested Solutions

INTRODUCTION

Problem 1 Aggregate Output. During a given year, the following activities occur:

- A silver mining company pays its workers \$200,000 to mine 75 pounds of silver. The silver is then sold to a jewelry manufacturer for \$300,000.
- The jewelry manufacturer pays its workers \$250,000 to make silver necklace, which the manufacturer sells directly to consumers for \$1,000,000.

1) Using the production-of-final goods approach, what is GDP in this economy?

Answer: All silver mined from the silver mining company was used to produce silver necklaces, so all silver is intermediate goods. So the value of final goods in this economy is \$1,000,000.

2) What is the value added at each stage of production? Using the value-added approach, what is GDP?

Answer: The silver mining company does not use intermediate goods. Its value added is simply equal to the value of the silver it mines, \$300,000. The jewelry company uses silver as an intermediate good. Thus, the value added by the jewelry company is equal to the value of the silver necklaces it manufactures minus the value of the silver it uses in production, \$1,000,000-\$300,000=\$700,000. Total value added in the economy equals \$300,000+\$700,000=\$1,000,000.

3) What are the total wages and profits earned? Using the income approach, what is GDP?

Answer: Of the \$300,000 of value added by the silver mining company, \$200,000 goes to worker (labor income) and the remaining \$100,000 goes to the firm (capital income). Of \$700,000 of value added by the jewelry manufacturer, \$250,000 goes to labor income and \$350,000 to capital income. For the economy as a whole, labor income is equal to \$450,000 (\$200,000+\$250,000) and capital income is equal to \$550,000 (\$100,000+\$450,000). From the income side, \$1,000,000.

You may find it helpful to fill in the table below:

Sliver Mining Company		Jewelry Manufacturer	
Revenues from sales	\$300,000	Revenues from sales	\$1,000,000
Expenses	\$200,000	Expenses	\$550,000
Wages	\$200,000	Wages	\$250,000
		Silver purchases	\$300,000
Profit	\$100,000	Profit	\$450,000

THE GOODS MARKET

Problem 1 Balanced Budget Multiplier. Consider the equilibrium condition in the goods market:

$$Y^* = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T].$$

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- 1) Suppose the government decides to increase T and G by one unit at the same time. What is the change in equilibrium output?

Answer: One unit increase in G leads to an increase in Y by $\frac{1}{1 - c_1}$ and one unit increase in T leads to a decrease in Y by $\frac{c_1}{1 - c_1}$. Thus, the equilibrium output changes by $\frac{1 - c_1}{1 - c_1} = 1$.

- 2) Based on your answer to 1), comment on the idea that “fiscal policy is macroeconomically neutral if balanced budget is maintained”.

Answer: Since balanced budget changes in G and T lead to one unit increase in equilibrium output, it is not macroeconomically neutral.

- 3) How will the change in the propensity to consume affect your answer to 1)? Why?

Answer: The change in propensity to consume does not affect the answer to 1). It is because the balanced tax increase offsets the multiplier effects of increase in G .

Problem 2 Automatic Stabilizer. *In the lecture we assume the fiscal policy to be exogenous. In reality, however, taxes typically depend on the level of the income and tend to be higher when income is higher. In this question we consider a case when this automatic response of taxes can help “stabilize” the economy when there is a change to autonomous spending.*

Consider the following equations:

$$C = c_0 + c_1(Y - T)$$

$$T = t_0 + t_1Y$$

$$Y = C + \bar{I} + G.$$

- 1) Solve for equilibrium output.

Answer: Equilibrium condition in the goods market requires that $Y = Z$

$$\begin{aligned} Y &= Z \\ &= C + I + G \\ &= c_0 + c_1(Y - T) + \bar{I} + G \\ &= c_0 + c_1(Y - t_0 - t_1Y) + \bar{I} + G \end{aligned}$$

Solving for Y , we get

$$Y^* = \frac{1}{1 - c_1(1 - t_1)}[c_0 + \bar{I} + G - c_1t_0]$$

- 2) What is the multiplier? How does the change of t_1 affect the multiplier? Explain your answer.

Answer: The multiplier in this economy is $\frac{1}{1 - c_1(1 - t_1)}$. As t_1 increase, the multiplier decreases, so does the equilibrium output.

3) Why is fiscal policy in this case called automatic stabilizer?

Answer: As long as $0 < t_1 < 1$, because of the automatic effect on taxes on the economy, the economy responds less to changes in autonomous spending than in the case where taxes are independent of income. Note that positive change in autonomous spending increases total taxes because of the increase in income, and this tends to lessen the increase in output. Conversely, a negative change in autonomous spending lowers total taxes because of the decrease in income, which tends to lessen the decrease in output. Since output tends to vary less and to be more stable, fiscal policy is called an automatic stabilizer.

Problem 3 Endogenous Investment. *This problem examines the implications of allowing investment to depend on output. IS-LM model carries this analysis much further and introduces an essential relation—the effect of the interest rate on investment—not examined in this problem.*

Suppose the economy is characterized by:

$$C = c_0 + c_1(Y - T)$$

$$I = b_0 + b_1Y$$

$$Y = C + I + G.$$

1) Solve for equilibrium output.

Answer: Equilibrium condition in the goods market requires that $Y = Z$

$$\begin{aligned} Y &= Z \\ &= C + I + G \\ &= c_0 + c_1(Y - T) + b_0 + b_1Y + G \end{aligned}$$

Solving for Y , we get

$$Y^* = \frac{1}{1 - (c_1 + b_1)}[c_0 + b_0 + G - c_1T]$$

2) What is the value of the multiplier? How does the relation between investment and output affect the value of the multiplier? For the multiplier to be positive, what condition must $(c_1 + b_1)$ satisfy? Explain your answers.

Answer: The multiplier in this economy is $\frac{1}{1 - (c_1 + b_1)}$. The parameter b_1 represents the coefficient of investment induced by income, showing how sensitive the investment reacts to the change in Y . As b_1 increases, the multiplier becomes larger. For the multiplier to be positive, $(c_1 + b_1)$ must to be smaller than 1.

3) What would happen if $(c_1 + b_1) > 1$?

Answer: Note that $(c_1 + b_1)$ represents the slope of the ZZ curve.

$$\begin{aligned} Z &= C + I + G \text{ by definition} \\ &= c_0 + c_1(Y - T) + b_0 + b_1Y + G \\ &= (c_1 + b_1)Y + [c_0 + b_0 + G - c_1T] \end{aligned}$$

Thus, if $(c_1 + b_1) > 1$, no equilibrium exists in this economy. Demand always exceeds the production and the economy explodes or diverges forever.

- 4) Suppose that the parameter b_0 , sometimes called *business confidence*, increases. How will equilibrium output be affected? Will investment change by more or less than the change in b_0 ? Why? What will happen to national saving?

Answer: Assume that b_0 changes by Δb_0 . This leads to the change in equilibrium output by $\frac{\Delta b_0}{1 - (c_1 + b_1)} > \Delta b_0$ because of the multiplier effect. This is because the changes in b_0 lead to the changes in autonomous spending, in turn, changes in output which lead to changes in income, and again, changes in demand, and so on. This chain reaction continues until the economy reaches a new equilibrium. Because of the increase in equilibrium output, the investment will increase, and so will national saving by the same amount as the increase in investment (since the economy is closed).

THE FINANCIAL MARKET

Problem 1 Money Demand vs. Bonds Demand. Suppose that a person's wealth is \$50,000 and that her yearly income is \$60,000. Also suppose that her money demand function is given by

$$M^d = \$Y(0.35 - i)$$

- 1) Derive the demand for bonds, which is denoted as B^d . Suppose the interest rate increases by 10 percentage points. What is the effect on her demand for bonds? (*Hint: Demand for money and demand for bonds must add up to her wealth.*)

Answer: Note that the economic agent is allocating her wealth into two types of financial assets, money, and bonds. Thus, demand for bonds is given by

$$\begin{aligned} B^d &= 50,000 - M^d \\ &= 50,000 - 60,000(0.35 - i) \\ &= 29,000 - 60,000i \end{aligned}$$

If i increases by 10 percentage points, her demand for bonds decreases by 6,000.

- 2) What are the effects of an increase in wealth on her demand for money and her demand for bonds? Explain in words.

Answer: While holding other variables fixed, if wealth increases, only her demand for bonds increases by the same amount as in the increased wealth. Note that the demand for money does not depend on wealth but on income.

- 3) What are the effects of an increase in income on her demand for money and her demand for bonds? Explain in words.

Answer: While holding other variables fixed, if income increases by ΔY , only her demand for money increases by $0.35\Delta Y$. Since her total wealth does not change, her demand for bonds decreases by $0.35\Delta Y$.

Problem 2 Money and Banking System. In this question, let us assume that

- The public holds no currency;
- The ratio of reserves to deposits is 0.1, and no bank holds excessive reserve;
- The demand for money is given by

$$M^d = \$Y(0.8 - 4i).$$

Initially, the supply of central bank money is \$100 billion, and nominal income is \$5 trillion.

- 1) What is the demand for central bank money H^d ? (*Hint: How is it related to M^d , when the public holds no currency and no bank holds excessive reserve?*)

Answer:

$$\begin{aligned} H^d &= \theta M^d \\ &= 0.1 \times \$5 \text{ trillion}(0.8 - 4i) \end{aligned}$$

- 2) Find the equilibrium interest rate by setting the demand for central bank money equal to the supply of central bank money.

Answer: Equilibrium condition in the central bank money market requires that $H^s = H^d$.

$$\$100 \text{ billion} = 0.1 \times \$5 \text{ trillion}(0.8 - 4i)$$

Solving for i , we get $i = 0.15$.

- 3) What is the overall supply of money? Is it equal to the overall demand for money at the interest rate you found in 2)?

Answer: Since $H^s = H^d = 0.1 \times M^d = M^s$, $M^s = 10 \times H^s$. Thus, the overall supply of money M^s is \$1,000 billion. Plugging the equilibrium interest rate $i = 0.15$ into the overall money demand equation to get

$$\begin{aligned} M^d &= \$5 \text{ trillion}(0.8 - 0.4 \times 0.15) \\ &= \$5 \text{ trillion}(0.8 - 0.6) \\ &= \$5 \text{ trillion}(0.2) \\ &= \$1 \text{ trillion}(= \$1,000 \text{ billion}). \end{aligned}$$

Hence, we have seen that the overall money supply is equal to the overall demand for money at the equilibrium interest rate.

- 4) What is the effect on the interest rate if central bank money is increased to \$300 billion?

Answer: If H^s increases, equilibrium interest rate decreases. By the equilibrium condition in the central bank money market,

$$\$300 \text{ billion} = 0.1 \times \$5 \text{ trillion}(0.8 - 4i)$$

Solving for i , we get $i' = 0.05$

- 5) If the overall money supply increases to \$3,000 billion, what will be the effect on i ?

Answer: If M^s increases, equilibrium interest rate decreases. By the equilibrium condition in the overall money market,

$$\$3,000 \text{ billion} = \$5 \text{ trillion}(0.8 - 4i)$$

Solving for i , we get $i' = 0.05$

THE $IS - LM$ MODEL

Problem 1 Numerical IS-LM. Consider the following numerical example of the $IS - LM$ model:

$$C = 200 + 0.25Y_D$$

$$I = 150 + 0.25Y - 1000i$$

$$G = 250$$

$$T = 200$$

$$\bar{i} = 0.05$$

- 1) Derive the IS relation.

Answer: Recall that the IS relation represents the equilibrium in the goods market, and it is plotted as a downward sloping curve on a $Y - i$ plane. Thus, equilibrium condition goods market is given by

$$\begin{aligned} Y &= C + I + G \\ &= 200 + 0.25(Y - 200) + 150 + 0.25Y - 1000i + 250 \end{aligned}$$

Rearrange the terms to get:

$$Y = 1100 - 2000i$$

- 2) The central bank sets an interest rate of 5%. How is that decision represented in the equations?

Answer: The decision of the central bank on interest rate is shown as a horizontal LM curve at $\bar{i} = 0.05$.

- 3) What is the level of real money supply when the interest rate is 5%? Use the expression:

$$\frac{M^d}{P} = 2Y - 8000i$$

Answer: By the equilibrium condition in the real money market, $\frac{M^s}{P} = \frac{M^d}{P}$.

$$\frac{M^s}{P} = 2Y - 8000i$$

At $\bar{i} = 0.05$,

$$\frac{M^s}{P} = 2Y - 400$$

- 4) Solve for the equilibrium values of C , and I , and verify the value you obtained for Y by adding C , I , and G .

Answer: At $\bar{i} = 0.05$, the goods market equilibrium is obtained at the intersection between IS curve and LM curve. Thus, from the IS curve derived in 1), $Y^* = 1100 - 2000 \times 0.05 = 1100 - 100 = 1000$. $C^* = 200 + 0.25(1000 - 200) = 200 + 0.25(800) = 200 + 200 = 400$. $I^* = 150 + 0.25 \times 1000 - 1000 \times 0.05 = 150 + 250 - 50 = 350$. We can verify that $C^* + I^* + G = 400 + 350 + 250 = 1000 = Y^*$.

- 5) Now suppose that government spending increases to $G = 400$. Explain the effects of an expansionary fiscal policy on Y , I , C , and real money supply.

Answer: An increase in government spending shifts the IS curve to the right. The new IS curve is derived by

$$\begin{aligned} Y &= C + I + G' \\ &= 200 + 0.25(Y - 200) + 150 + 0.25Y - 1000i + 400 \end{aligned}$$

Rearranging terms,

$$Y = 1400 - 2000i$$

At $\bar{i} = 0.05$, $Y_1 = 1400 - 2000 \times 0.05 = 1400 - 100 = 1300$. $C_1 = 200 + 0.25(1300 - 200) = 200 + 275 = 475$. $I_1 = 150 + 0.25 \times 1300 - 1000 \times 0.05 = 150 + 325 - 50 = 425$. $\left(\frac{M^s}{P}\right)_1 = 2 \times 1300 - 400 = 2200$. An expansionary fiscal policy increases equilibrium output, consumption, investment, and real money supply.

- 6) Starting from an interest rate 5% and government spending equal to 250 units, increase government spending to 400 units while fixing the real money supply at 1600 units. Compare the effect of the increase in the government spending on Y , I , and C in 5) and explain the difference.

Answer: Note that once government spending increases, the real money demand also increases because of an increase in equilibrium output. However, if the real money supply is fixed at 1600, interest rate must go up. Thus, the interest rate becomes to be dependent on nominal income and the real money supply.

Rewriting i as a function of the real money supply and nominal income,

$$i = \frac{2Y - 1600}{8000}$$

Deriving IS curve,

$$\begin{aligned} Y &= C + I + G \\ &= 200 + 0.25(Y - 200) + 150 + 0.25Y - 1000i + 400 \\ &= 200 + 0.25(Y - 200) + 150 + 0.25 - 1000 \left(\frac{2Y - 1600}{8000} \right) + 400 \end{aligned}$$

Solving for Y , $Y_2 = 1200$ ($< Y_1 = 1300$). Computing remaining variables, $C_2 = 450$ ($< C_1 = 475$), $I_2 = 350$ ($< I_1 = 425$).

Note that $i_2 = \frac{2 \times 1200 - 1600}{8000} = \frac{800}{8000} = 0.1$. An increase in government spending increases the equilibrium output, consumption, and investment, but with a smaller amount than the case where the real money supply increases. It is because the interest rate must increase while fixing the real money supply, leading to a smaller increase in investment which, in turn, leads to a smaller increase in output, a smaller increase in income.

Now return to the initial condition in which government spending $G = 250$. Suppose that the central bank cuts the interest rate to 3%.

- 7) How does this change the LM curve? Solve for the equilibrium values for Y , I , C , and real monetary supply. Explain in words the effects of an expansionary monetary policy.

Answer: The LM curve shifts to the downward. Going back to the original IS curve, $Y = 1100 - 2000i$. Plugging $i = 0.03$ into the original IS curve, solving for Y , and computing the remaining variables, we get

$$\begin{aligned} Y_3 &= 1040 (> Y^* = 1000) \\ C_3 &= 410 (> C^* = 400) \\ I_3 &= 380 (> I^* = 350) \\ \left(\frac{M^s}{P} \right)_3 &= 1840 (> \left(\frac{M^s}{P} \right)^* = 1600). \end{aligned}$$

An expansionary monetary policy increases the equilibrium output, consumption, investment, and real money supply.

THE FINANCIAL MARKET II

Problem 1 The Troubled Asset Relief Program (TARP). Consider a bank that has assets of 100, capital of 20, and short term credit of 80. Among the bank's assets are securitized assets whose value depends on the price of houses. These assets have a value of 50.

- 1) Set up the bank's balance sheet.

Answer:

Assets	Liabilities
\$100	Short term credit: \$80
(securitized assets: \$50	
+ the other assets: \$50)	Capital: \$20

Suppose that as a result of a housing price decline, the value of the bank's securitized assets falls by an uncertain amount, so that these assets are now worth somewhere between 25 and 45. Call the securitized assets "troubled assets." The value of the other assets remains at 50. As a result of the uncertainty about the value of the bank's assets, lenders are reluctant to provide any short-term credit to the bank.

- 2) Given the uncertainty about the value of the bank's assets, what is the range in the value of the bank's capital?

Answer: The value of the bank's capital ranges from $-\$5$ to $\$15$.

As a response to this problem, the government considers purchasing the troubled assets, with the intention of reselling them again when the markets stabilize. (This is the original version of the TARP.)

- 3) If the government pays 25 for the troubled assets, what will be the value of the bank's capital? How much would the government have to pay for the troubled assets to ensure that the bank's capital does not have a negative value? If the government pays 45 for the troubled assets, but the true value turns out to be much lower, who bears the cost of this mistaken valuation? Explain.

Answer: The purchase of troubled assets by the government does not change the bank's capital. To ensure that the bank's capital does not have a negative value, the government has to pay minimum 30 for the troubled assets. If the government pays 45 for the troubled assets, but the true values turns out to be much lower, the government bears the cost of the mistaken valuation.

Suppose instead of buying the troubled assets, the government provides capital to the bank by buying ownership shares, with the intention of reselling the shares when the markets stabilize. (This is what the TARP ultimately became.) The government exchanges treasury bonds (which become assets for the bank) for ownership shares.

- 4) Suppose the government exchanges 25 of Treasury bonds for ownership shares. Assuming the worst-case scenario (so that the troubled assets are worth only 25), set up the new balance sheet of the bank. (Remember that the firm now has three assets: 50 of untroubled assets, 25 of troubled assets, and 25 of Treasury bonds.) What is the total value of the bank's capital? Will the bank be insolvent?

Answer:

Assets	Liabilities
\$100	Short term credit: \$80
(troubled assets: \$25	
+ the other assets: \$50	
+ treasury bonds: \$25)	Capital: \$20

The total value of the bank's capital is \$20. The bank will be solvent.

- 5) Given your answers and the material in the text, why might recapitalization be a better policy than buying the troubled assets?

Answer: Recapitalization might be a better policy than buying troubled assets because the former directly affects capital, but the latter does not.

THE MEDIUM RUN

Problem 1 The Natural Rate of Unemployment. Suppose that the markup of goods prices over marginal cost is 5%, and that the wage-setting equation is

$$W = P(1 - u)$$

where u is the unemployment rate.

- 1) What is the real wage, as determined by the price-setting equation?

Answer: Recall that the price-setting equation says

$$\frac{W}{P} = \frac{1}{1 + m}$$

Plugging $m = 0.05$ into the above equation, we get

$$\frac{W}{P} = \frac{1}{1 + 0.05} \approx 0.95$$

- 2) Solve for the natural rate of unemployment.

Answer: Dividing both sides of the wage-setting equation by P to get in terms of real wage,

$$\frac{W}{P} = 1 - u$$

Note that the natural rate of unemployment is derived at the intersection of WS equation and PS equation. Thus,

$$\begin{aligned} 1 - u_n &= \frac{1}{1 + 0.05} \approx 0.95 \\ u_n &\approx 0.05 \text{ (5\%)} \end{aligned}$$

- 3) What happens to the natural rate of unemployment if the markup of prices over costs increases to 10%? Explain the logic behind your answer.

Answer: If markup increases to 10%, the real wage decreases to $\frac{1}{1 + 0.1} \approx 0.91$. The natural rate of unemployment increases to $1 - u_n \approx 0.91$, $u_n \approx 0.09$ (9%). The increase in markup shifts the PS curve to the downward, leading to an increase in the natural rate of unemployment.

Problem 2 Mutations of the Phillips Curve. Suppose that the Phillips curve is given by

$$\pi_t = \pi_t^e + 0.1 - 2u_t$$

and expected inflation is given by

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

and suppose that θ is initially equal to 0 and $\bar{\pi}$ is given and does not change. It could be zero or any positive value. Suppose that the rate of unemployment is initially equal to the natural rate. In year t , the government decides to bring the unemployment rate down to 3% and hold it there forever.

- 1) Determine the rate of inflation in periods $t, t + 1, t + 2, t + 3, t + 4, t + 5$.

Answer: Note that initially, $\theta = 0$, so $\pi_t^e = \bar{\pi}$ and the rate of unemployment is equal to the natural rate, which means that $\pi_t - \pi_t^e = 0.1 - 2u_n = 0$. Thus, $\pi_t = \bar{\pi}$. From the year $t + 1$, the unemployment rate reduces to 3%.

$$\pi_{t+1} = \bar{\pi} + 0.1 - 2 \times 0.03 = \bar{\pi} + 0.04$$

$$\pi_{t+2} = \bar{\pi} + 0.1 - 2 \times 0.03 = \bar{\pi} + 0.04$$

⋮

$$\pi_{t+5} = \bar{\pi} + 0.1 - 2 \times 0.03 = \bar{\pi} + 0.04$$

Inflation rate remains constant at $(\bar{\pi} + 0.04)$ from year $t + 1$ to $t + 5$.

- 2) Do you believe the answer given in 1)? Why or why not? (*Hint: Think about how people are more likely to form expectations of inflation.*)

Answer: The answers obtained in 1) do not seem plausible. If the inflation consistently deviates from the target rate, wage setters would eventually update their expectation formation rule.

Now suppose that in year $t + 6$, θ increases from 0 to 1. Suppose the government still decides to keep the unemployment at 3% forever.

- 3) Why might θ increase in this way?

Answer: When θ increase to 1, $\pi_t^e = \pi_{t-1}$, which means that the expectation of inflation rate is de-anchored. As people update their expectation rule, so the previous inflation rate persists in the economy.

- 4) Calculate the inflation rate in years $t + 6, t + 7$, and $t + 8$.

Answer:

$$\pi_{t+6} = \pi_{t+5} + 0.1 - 2 \times 0.03 = \pi_{t+5} + 0.04 = \bar{\pi} + 0.04 + 0.04 = \bar{\pi} + 0.08$$

$$\pi_{t+7} = \pi_{t+6} + 0.1 - 2 \times 0.03 = \pi_{t+6} + 0.04 = \bar{\pi} + 0.08 + 0.04 = \bar{\pi} + 0.12$$

$$\pi_{t+8} = \pi_{t+7} + 0.1 - 2 \times 0.03 = \pi_{t+7} + 0.04 = \bar{\pi} + 0.12 + 0.04 = \bar{\pi} + 0.16$$

- 5) What happens to inflation when $\theta = 1$ and unemployment is kept below the natural rate of unemployment?

Answer: When $\theta = 1$ and unemployment is kept below the natural rate of unemployment, the change in inflation is positive each year and inflation continues to increase as in 4).

- 6) What happens to inflation when $\theta = 1$ and unemployment is kept at the natural rate of unemployment?

Answer: When $\theta = 1$ and unemployment is kept at the natural rate of unemployment, the change in inflation is zero. So inflation is constant and equal to the lagged inflation.

THE LONG RUN

Problem 1 The Cobb-Douglas Production Function and the Steady State. Suppose that the economy's production function is given by

$$Y = K^{1/3}N^{2/3}$$

- 1) Transform the production function into a relation between output per worker and capital per worker.

Answer: Dividing both sides of the production function by N ,

$$\begin{aligned}\frac{Y}{N} &= K^{1/3}N^{-1/3} \\ &= \left(\frac{K}{N}\right)^{1/3} \\ \Rightarrow y &= k^{1/3} = f(k)\end{aligned}$$

where $y = \frac{Y}{N}$ denotes the output per worker, $k = \frac{K}{N}$ capital per worker.

- 2) For a given saving rate s , and depreciation rate δ , derive an expression for capital per worker and output per worker in the steady state.

Answer: Recall that the capital accumulation equation is given by

$$K_{t+1} - K_t = sF(K_t, N_t) - \delta K_t$$

Dividing both sides by N_t ,

$$k_{t+1} - k_t = sf(k_t) - \delta k_t$$

By definition of steady state, steady state capital is obtained at $k_{t+1} - k_t = 0$. Thus, solving for k_t ,

$$\begin{aligned}sf(k_t) &= \delta k_t \\ sk_t^{1/3} &= \delta k_t \\ \frac{s}{\delta} &= k_t^{2/3} \\ \Rightarrow k^* &= \left(\frac{s}{\delta}\right)^{3/2} \\ y^* = f(k^*) &= (k^*)^{1/3} = \left(\left(\frac{s}{\delta}\right)^{3/2}\right)^{1/3} = \left(\frac{s}{\delta}\right)^{1/2}\end{aligned}$$

- 3) Solve for the steady-state level of output per worker when $s = 0.32$ and $\delta = 0.08$.

Answer: When $s = 0.32, \delta = 0.08$, $y^* = \left(\frac{0.32}{0.08}\right)^{1/2} = 4^{1/2} = 2$.

- 4) Briefly explain about the golden level of consumption. Suppose that the depreciation rate remains constant at $\delta = 0.08$. Solve for the golden-rule saving rate, golden-rule level of capital per worker, golden rule level of output per worker, and golden-rule level of consumption.

Answer: The golden level of consumption refers to the steady-state level of consumption where it is maximized. The golden rule of steady state is derived by solving the following problem:

$$\begin{aligned} \max_s c &= \frac{C}{N} = (1-s)\frac{Y}{N} \\ \text{s.t. } sf(k) &= \delta k \end{aligned}$$

Note that we can rewrite the objective function as $c = (1-s)y = (1-s)f(k) = (1-s)k^{1/3}$. Implied by the constrain, we know that $k^* = \left(\frac{s}{\delta}\right)^{3/2}$. Plugging k^* into the given objective function,

$$\begin{aligned} \max_s c &= (1-s) \left(\left(\frac{s}{\delta} \right)^{3/2} \right)^{1/3} \\ \Rightarrow \max_s c &= (1-s) \left(\frac{s}{\delta} \right)^{1/2} \end{aligned}$$

Since the objective function is concave in s , taking first order condition with respect to s , we get

$$\frac{\partial c}{\partial s} = \frac{1}{\delta} \left(-s^{\frac{1}{2}} + \frac{1}{2}(1-s)s^{-\frac{1}{2}} \right) = 0$$

Solving for s , $s_{gr} = \frac{1}{3} \approx 0.33$. Computing the remaining variables,

$$\begin{aligned} k_{gr} &= \left(\frac{s_{gr}}{\delta} \right)^{3/2} = \left(\frac{0.33}{0.08} \right)^{3/2} \approx 4.125^{3/2} \approx 8.38 \\ y_{gr} &= \left(\left(\frac{s_{gr}}{\delta} \right)^{3/2} \right)^{1/3} = \left(\frac{s_{gr}}{\delta} \right)^{1/2} = \left(\frac{0.33}{0.08} \right)^{1/2} \approx 2.03 \\ c_{gr} &= (1-s_{gr})y_{gr} \approx 0.67 \times 2.03 \approx 1.36 \end{aligned}$$

- 5) Suppose that the depreciation rate remains constant at $\delta = 0.08$, while the saving rate is reduced to $s = 0.16$. What is the new steady-state output per worker?

Answer: If s changes to $s' = 0.16$,

$$\begin{aligned} k' &= \left(\frac{s'}{\delta} \right)^{3/2} = \left(\frac{0.16}{0.08} \right)^{3/2} = 2^{3/2} \\ y' &= (k')^{1/3} = (2^{3/2})^{1/3} = 2^{1/2} = \sqrt{2} \approx 1.4 \end{aligned}$$

Problem 2 Growth Accounting. Assume the production function is given by

$$Y = K^{1/3}(AN)^{2/3}$$

- 1) Show that the rate of growth of total factor productivity $g_A = \frac{3}{2}(g_Y - g_N) - \frac{1}{2}(g_K - g_N)$ where g_Y denotes the growth rate of Y , g_K the growth rate of capital, and g_N growth rate of the labor input.

Answer: Dividing both sides of the production function by N , we have

$$\frac{Y}{N} = A^{2/3} \left(\frac{K}{N} \right)^{1/3}$$

Take log on both sides to get

$$\log Y - \log N = \frac{2}{3} \log A + \frac{1}{3}(\log K - \log N)$$

The above equation implies that $g_Y - g_N = \frac{2}{3}g_A + \frac{1}{3}(g_K - g_N)$. Rearrange terms to get

$$\begin{aligned} 3(g_Y - g_N) &= 2g_A + (g_K - g_N) \\ \Rightarrow \frac{3}{2}(g_Y - g_N) &= g_A + \frac{1}{2}(g_K - g_N) \\ \Rightarrow g_A &= \frac{3}{2}(g_Y - g_N) - \frac{1}{2}(g_K - g_N) \end{aligned}$$

- 2) What do the quantity $g_Y - g_N$ and $g_K - g_N$ represent, respectively?

Answer: The quantity $g_Y - g_N$ represents the growth rate of output per worker and the quantity $g_K - g_N$ the growth rate of capital per worker, respectively.